

Advanced PlayStation Audio '98



Libspu optimizations
Introduction to libmidi



libspu - Volume Modes

- SPU master volume and voice volume
 - Range -0x4000 to 0x3fff
- CD input
 - Range -0x8000 to 0x7fff
- Direct volume mode
 - Volume is set to input parameter
 - Negative volumes allow for 3D Sound
 - X360

libspu - Volume Modes

- Exponential & linear volume modes
 - Increase
 - Decrease
 - Normal phase
 - Reverse phase (for negative volumes)
 - Ramps from current level to 0 or maximum volume in hardware

libspu - Volume Modes

- Exponential & linear volume modes
 - Parameter determines ramp speed
 - 0 is instantaneous (or nearly so)
 - Below $\sim 0x28$, an inverse value is added per tick
 - $\sim 0x3f$ increases the volume 4-7 points per 16 ticks
 - $> \sim 0x3f$, the same volume increase takes more ticks
 - $0x7f$ is infinite (or nearly so)
 - Verify correct mode-sign combination

libspu - Volume Modes

- Exponential & linear volume modes
 - `SpuGetVoiceVolume()` returns input parameters
 - `SpuGetVoiceVolumeX()` returns actual volume
 - Scales volumes from -0x8000 to 0x7fff
 - Can stop an effect very near a desired point
 - Set a timer
 - `SpuGetVoiceVolume(voice, &voll, &volr )`
 - `SpuSetVoiceVolumeAttr(voice, voll>>1, volr>>1, SPU_VOICE_DIRECT, SPU_VOICE_DIRECT)`

libspu - Voice Allocation Tips

- Methods of testing voice status
 - SpuGetKeyStatus()
 - SpuGetAllKeysStatus()
 - SpuRGetAllKeysStatus()
- Components of voice status
 - Software key on/off
 - Envelope playback

libspu - Voice Allocation Tips

- Voice status
 - SPU_ON
 - Keyed on, voice actively playing
 - SPU_OFF
 - Keyed off in software, voice finished playing
 - SPU_ON_ENV_OFF
 - Keyed on, voice not yet playing OR
 - Keyed on, one shot VAG expired
 - SPU_OFF_ENV_ON
 - Keyed off in software, release state entered

libspu - Voice Allocation Tips

- Possible problems
 - SPU_ON_ENV_OFF state ambiguous
 - SPU registers only flushed every 1/44,000 second
 - Wasted cycles occur if voice status checked every allocation
- Solution - create a new voice status check and voice allocation scheme

libspu - Voice Allocation Tips

- Set in key on/off functions
 - Voice on/off flag (vo_flag)
 - Tracks software key on/off
 - Newly keyed on flag (nko_flag)
 - Flags voice for first frame of key on
 - Looping voice flag (lp_flag)
 - Sound data structure must include loop info

libspu - Voice Allocation Tips

- Set once per frame (or less)
 - Voice free flag (vf_flag)
 - Is the voice allocable?
 - Envelope on/off flag (env_flag)
 - Is the voice playing data?
 - One-shot expiry flag (exp_flag)
 - Has one-shot voice naturally expired?
 - nko_flag and exp_flag combine to resolve SPU_ON_ENV_OFF ambiguity

libspu - Voice Allocation Tips

- Once per frame (or less often)
 - Push key on/off queue
 - Skip lp_flag low & nko_flag low voices
 - Check envelope state
 - Fill env_flag using SpuGetVoiceEnvelope()
 - If env_flag low & vo_flag high
 - Set exp_flag high OR
 - If exp_flag already high
 - Clear exp_flag & Clear vo_flag
 - Determine vf_flag with env_flag & vo_flag
 - Clear nko_flag & exp_flag

libspu - Keying Sounds On/Off

- Problems
 - Key on/off commands may be missed
 - Key on/off queue contains high bits for voices which must be keyed on or off
 - Old queue is flushed out
 - SPU registers only read every 1/44,000 second
 - Cycles wasted
 - Voices reallocated before playback begins
- Solution - queue ALL key on/off commands

libspu - Keying Sounds On/Off

- Create two additional 24-bit variables
 - keyoff_flag & keyon_flag
- In key on function
 - Bit OR keyon_flag & clear keyoff_flag bit
 - Bit OR nko_flag
 - Bit OR lp_flag if looping sound
 - Do NOT call SpuSetKey()

libspu - Keying Sounds On/Off

- In key off function
 - Bit OR keyoff_flag & clear keyon_flag bit
 - Clear nko_flag bit for that voice
 - Clear lp_flag bit for that voice
 - Do NOT call SpuSetKey()
- Once per frame, before voice status check
 - SpuSetKey(SPU_ON, keyon_flag)
 - SpuSetKey(SPU_OFF, keyoff_flag)

libmidi - Increased Functionality

- Channel-level pitch bend changes
 - Notes may be bent INTO
- RPN data
 - RPN 0 (pitch bend max and min setting)
 - RPN 1 (tone shift changes)
 - RPN 2 (tone center changes)
- Damper changes on a channel level, rather than a global level
- Reset all controllers supported

libmidi - Increased Functionality

- Song level pan shift/time effect
 - Similar to a crescendo/ decrescendo
- Voice level note shift/time effect
 - Doppler type effect
- Current tempo of songs retrievable
- Cancellation of transfer to SPU RAM
 - SsVabTransBodyPartly() does not support this
- Vibrato
- Portamento

libmidi - Increased Functionality

- Nested song loops
- Current tempo retrievable
- Cancellation of time-based effects
 - Reports back the amount of effect and time remaining
 - Volume, pan, tempo (song level only), pitch (voice level only)

libmidi - New File Formats

➤ MAB

- Reduced structure sizes
 - VabHdr level
 - ProgAtr level
 - VagAtr level
 - Vag Offset Table
 - Unused tone slots not stored

➤ MEQ

- Loop marker & callback marker messages reduced to one message
- File size reduced and MIDI time conserved

libmidi - Linkable Functions

- SPU RAM management
 - Use libspu standard SPU RAM management functions OR
 - Create smaller or “best-fit” functions OR
 - Hardcode all transfers to minimize code size
- Pitch calculation
 - Use library standard function OR
 - Create larger PLU table for more precision or smaller PLU table for less code size

libmidi - Linkable Functions

- Volume calculation
 - Least complex
 - Requested level volume and pan
 - Medium complexity
 - Requested and tone level of volume and pan
 - Exponential left/right separation of song voices
 - Most complex
 - Requested, tone, program, and bank levels of volume and pan
 - Exponential left/right separation of song voices

libmidi - Linkable Functions

- Voice allocation
 - First available voice & priority-based system
 - Addition of envelope size & age
- Low-level MIDI functions

More libmidi

- Reduced structure sizes
 - Song management table
 - Voice management
- Reduced tick callback cycle usage
 - Not wrapped around libspu calls
 - Data transfer to SPU registers only if necessary
 - Fewer I-cache misses at tick callback
- Standardization
 - Key on commands take volume and pan
 - Easier to pan voices
 - Seq/Sep duality removed - extra wrapper

libmidi - Tools

- Initially
 - seq2meq
 - VAB2MAB
- Coming
 - smf2meq
 - MABmaker